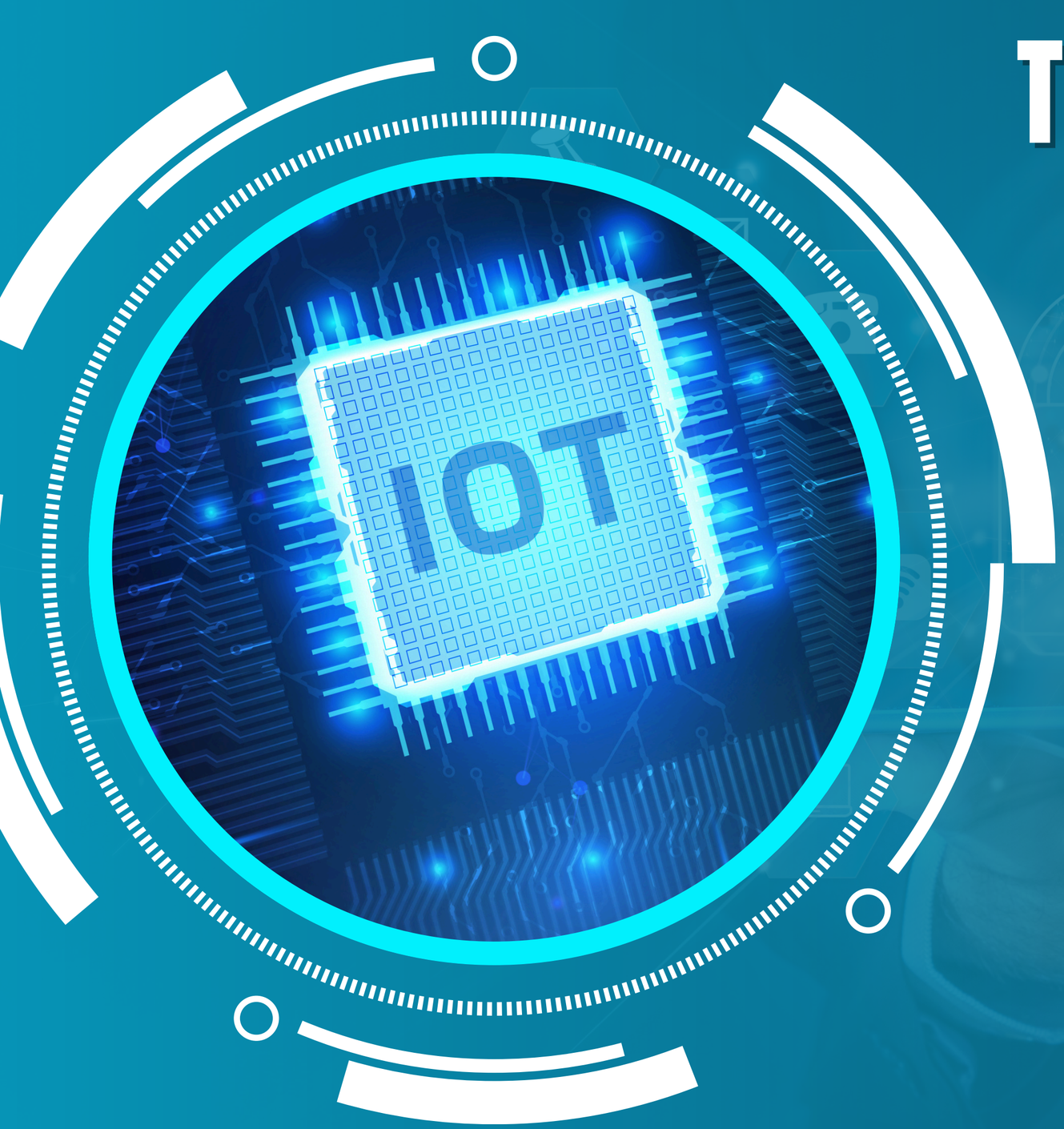
Digital Control System

SMART ENERGY-SAVING AND POWER MONITORING MODULE



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Introduction

PROJECT TITLE:

Smart Energy-Saving and Power Monitoring Module

DESCRIPTION :

Our IoT solution enables real-time power monitoring, error detection, and energy optimization for homes. Users control devices via a mobile app, ensuring efficient power-saving and enhanced safety.

Problem Statement

- No real-time visibility of energy usage.
- Manual monitoring is inconvenient.
- Higher bills due to wastage.
- Faulty devices remain unnoticed.

Objectives

Fault detection & automatic shutdown.

Real-time monitoring of power usage.

Mobile app for remote control.

Monthly cost prediction.

Features of the Solution

- Real-time monitoring dashboard.
- Automatic shutdown of faulty devices.
- Cost prediction using ML.
- Remote device control.

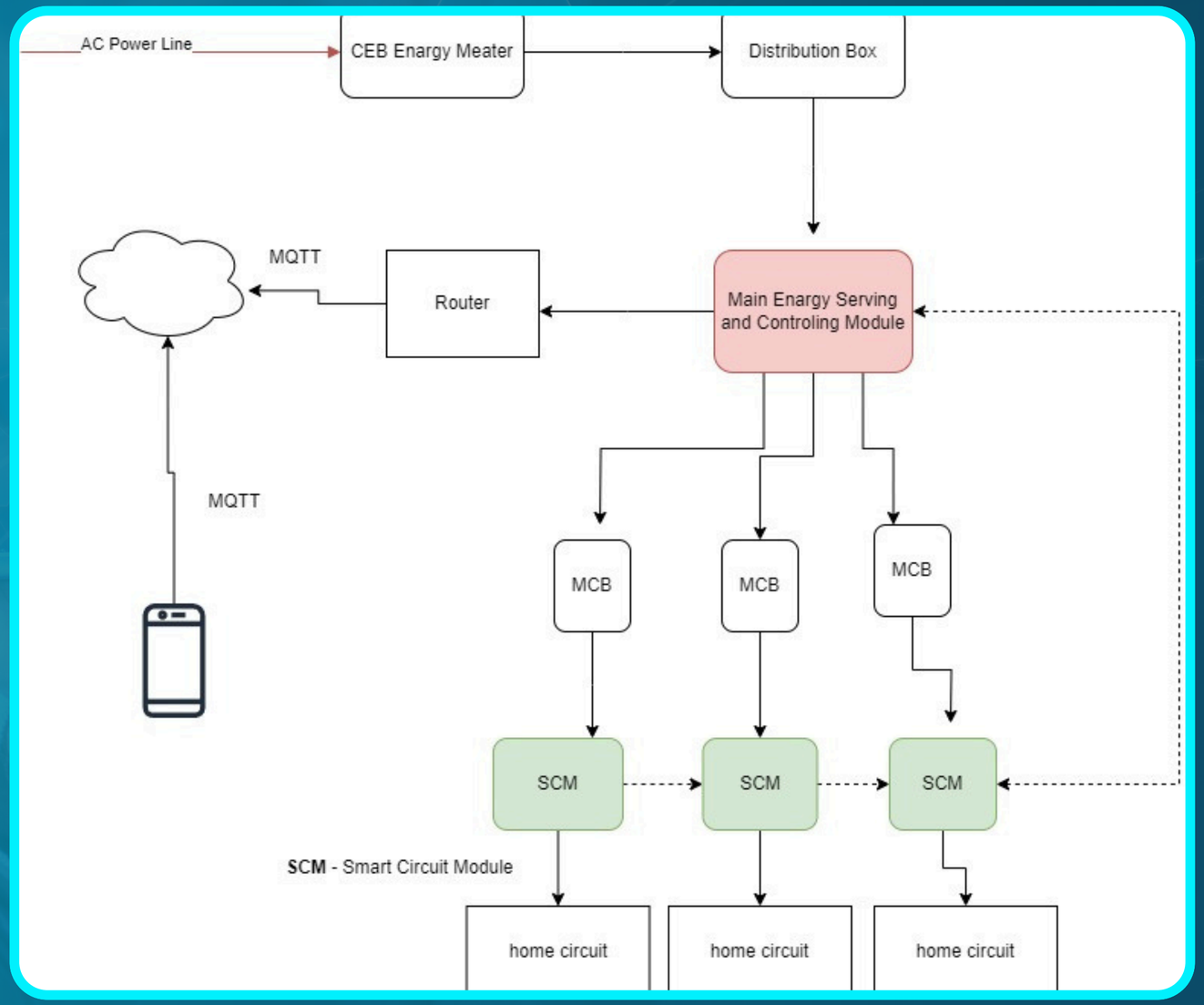
Methodology

Hardware:

- ESP32-s3
- pzem-004
- lcd display
- relay model
- temprature sensor

Software:

- Mobile app
- web application
- cloud storage.



Benefits

"Our Smart Energy-Saving and Power Monitoring Module offers multiple benefits for households. It ensures energy efficiency by reducing wastage and optimizing power usage, which directly contributes to cost savings on electricity bills."

Safety Enhancement

Protects the home by detecting and shutting down faulty devices.

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Protects the home by detecting and shutting down faulty devices.

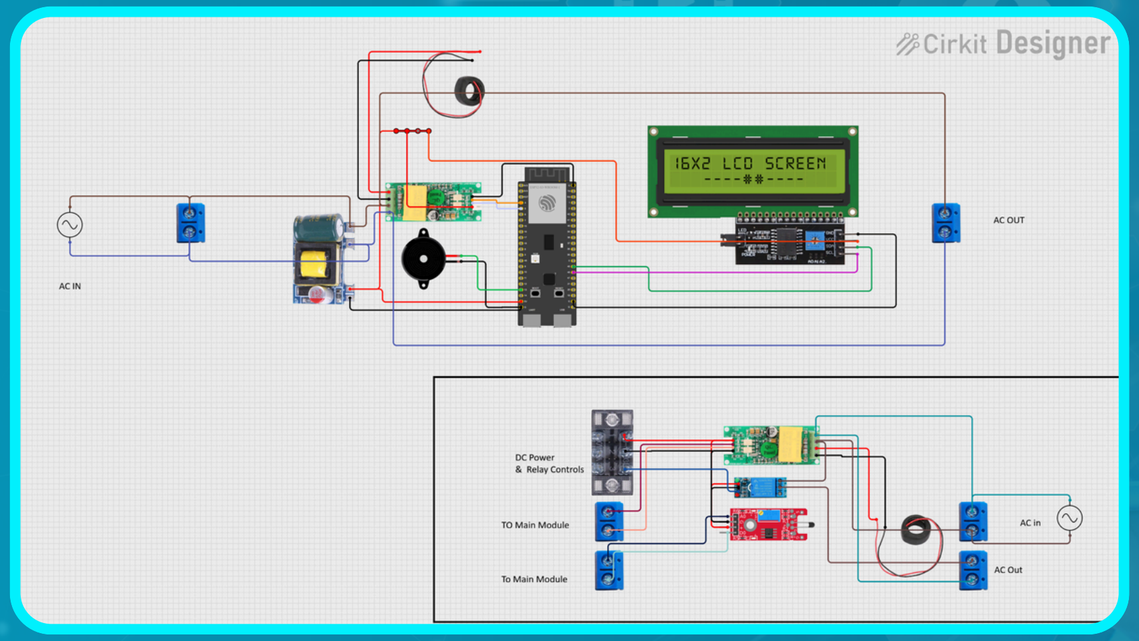
Cost Savings

Lowers electricity bills through smarter consumption and predictions.

Cost Savings

Provides easy control of appliances anytime via mobile app.

Project Budget



Component	Quantity	Unit Cost (LKR)	Total (LKR)
ESP32-S3 Microcontroller	1	1,850.00	1,850.00
PZEM Power Sensor Module	1	5,200.00	5,200.00
Relay Module	1	150	150
Buzzer	1	60	60
LCD 128x64 Display	1	1,640.00	1,640.00
I2C Module	1	150	150
Temperature Sensor Module	1	150	150

Total Estimated Cost: **LKR 9,200.00**

Our Team

We’re a team of students creating a smart IoT solution

SENITH
Team Member

ISHAN
Team Member

HASINTHA
Team Leader

SADEEPA
Team Member

DINETH
Team Member

SAVINDU
Team Member

Thank You for Attention

END OF THE PRESENTATION